EDS223 - HW2

Exploring patterns of environmental justice

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Rubric

Data analysis

- Code must produce expected output based on correct data manipulation
- Spatial operations must be performed correctly with appropriate verification (e.g., CRS matching, appropriate join methods, checking results)
- Multiple approaches can be used to reach the correct output. Analysis must demonstrate critical interrogation of the approach used by showing justification and verification of intermediate steps. "Correct" answers are not sufficient. Justification and verification of approach should be demonstrated using the following:
  - custom warning and error message (e.g. warning() and stop(); resources from EDS 221)
  - informative comments (resource from EDS 220)
  - BONUS: unit tests (e.g. using {testthat}; resources from EDS 221)

Plots, tables, and maps

- All plots, tables, and maps must be clear, accurate, and effectively convey the intended information
- All plots and maps must include the following elements:
  - an informative title
  - legends with legible titles, including units
  - color scales that are accessible (i.e. make intuitive sense) and appropriate to the data (i.e. discrete vs. continuous)
  - for maps: indication of scale and orientation (i.e. graticules/gridlines or scale bar and compass), and an appropiate basemap included where necessary
- All tables should be rendered with legible titles and styling (e.g. {gt})

Written reflections

- Reflections must be clear, accurate, and demonstrate a deep understanding of the analysis performed.

Professional Output

- The rendered Quarto doc must show all required elements in a professional style with explanation of each step of analysis
  - see examples of professional and unprofessional output on the Assignments page

Setup

```
# Load packages
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.3     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.2
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to beco
```

```
library(here)
```

```
here() starts at /Users/smsoriano/Documents/Documents - Nawiliwili/UCSB_MEDS/eds223/EDS223-
HW2
```

```
library(sf)
```

```
Linking to GEOS 3.13.0, GDAL 3.8.5, PROJ 9.5.1; sf_use_s2() is TRUE
```

```
library(tmap)
```

```
Registered S3 method overwritten by 'stars':
  method                   from
  st_interpolate_aw.stars sf
```

```
library(spData)
library(knitr)
library(ggpubr)

# Set warnings to display as they occur
options(warn = 1)

# Set ggplot theme
set_theme(theme_bw())
```

```
# Load data
# EJScreen data - environmental and demographic US info at the block group level
ejscreen <- sf::st_read(here("data","ejscreen",
                             "EJSCREEN_2023_BG_StatePct_with_AS_CNMI_GU_VI.gdb"))
```

```
Reading layer `EJSCREEN_StatePctiles_with_AS_CNMI_GU_VI' from data source
  `/Users/smsoriano/Documents/Documents - Nawiliwili/UCSB_MEDS/eds223/EDS223-
HW2/data/ejscreen/EJSCREEN_2023_BG_StatePct_with_AS_CNMI_GU_VI.gdb'
  using driver `OpenFileGDB'
Simple feature collection with 243021 features and 223 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: -19951910 ymin: -1617130 xmax: 16259830 ymax: 11554350
Projected CRS: WGS 84 / Pseudo-Mercator
```

```
# HOLC Redlining data - HOLC maps and area descriptions for LA
holc <- sf::st_read(here("data","mapping-inequality",
                             "mapping-inequality-los-angeles.json")) |>
        # Correct invalid geometries to avoid error in st_join when using CRS 4326
        st_make_valid()
```

```
Reading layer `mapping-inequality-los-angeles' from data source
  `/Users/smsoriano/Documents/Documents - Nawiliwili/UCSB_MEDS/eds223/EDS223-
HW2/data/mapping-inequality/mapping-inequality-los-angeles.json'
```

```
  using driver `GeoJSON'
Simple feature collection with 417 features and 14 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: -118.6104 ymin: 33.70563 xmax: -117.7028 ymax: 34.30388
Geodetic CRS:  WGS 84
```

```r
# Bike + Ped Crash data - LA crash records
crashes <- read_csv(here("data","SWITRS_LA","Crashes.csv")) |>
    # Remove rows w/ missing coordinates so convert to sf is possible
    # Use POINT_X,POINT_Y each with only 3% missing data (Lat/Long each have 38%)
    drop_na(POINT_X,POINT_Y) |>
    # Convert crashes csv to sf data - SWITRS uses CRS WGS 1984 EPSG:4326
    st_as_sf(coords = c("POINT_X","POINT_Y"), crs = 4326)
```

```
Rows: 251100 Columns: 80
-- Column specification --------------------------------------------------------
Delimiter: ","
chr  (45): COLLISION_TIME, OFFICER_ID, REPORTING_DISTRICT, CHP_BEAT_TYPE, CI...
dbl  (33): CASE_ID, ACCIDENT_YEAR, JURIS, DAY_OF_WEEK, CHP_SHIFT, POPULATION...
date  (2): PROC_DATE, COLLISION_DATE

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
# Warning - CRS - confirm crs match, transform if not
if ((st_crs(ejscreen)$epsg == st_crs(holc)$epsg) &
     (st_crs(ejscreen)$epsg == st_crs(crashes)$epsg) &
     (st_crs(holc)$epsg == st_crs(crashes)$epsg)){
    # All three CRS match - true
    print("CRS do match!")
} else {
    # CRS match - false
    warning("CRS do NOT match!")
    print("Changing EJScreen data CRS to match HOLC and Crashes data CRS EPSG:4326...")
    # CRS transforms to match ejscreen data
    ejscreen <- st_transform(ejscreen, crs = st_crs(holc)$epsg)
    crashes <- st_transform(crashes, crs = st_crs(holc)$epsg)
    print("CRS match complete!")
}
```

```
Warning: CRS do NOT match!
```

```
[1] "Changing EJScreen data CRS to match HOLC and Crashes data CRS EPSG:4326..."
[1] "CRS match complete!"
```

Part 1: Legacy of redlining in current environmental (in)justice

Question 1: Map of Historical Redlining Neighborhoods

```
# Correct HOLC colors per grade
holc_colors = c("A" = "#76a865",
                "B" = "#7cb5bd",
                "C" = "#f1db11",
                "D" = "#d9838d",
                "NA" = "#000000")
# Create holc polygon map over la county basemap
tm_basemap(server = "Esri.WorldTopoMap") +
tm_shape(holc) +
    tm_polygons(fill = "grade",
                fill_alpha = 0.5,
                lwd = 0.5,
                fill.scale = tm_scale_categorical(values = holc_colors),
                fill.legend = tm_legend(title = "HOLC Grade",
                                        # Specify legend layout
                                        position = tm_pos_out("right"))) +
tm_compass(position = c("left","bottom")) +
tm_scalebar(position = c("left","bottom")) +
tm_options(component.autoscale = FALSE) +
tm_title("Los Angeles County: Redlined Neighborhoods",
            frame = TRUE,
            size = 1.5) +
tm_layout(# Specify legend title size
          legend.title.size = 1.0,
          # Specify legend text size
          legend.text.size = 0.8,
          # Set font family
          text.fontfamily = "sans",
          # Set inner margins to give scalebar room
          inner.margins = c(0.1,0.1,0.05,0.05)
          )
```

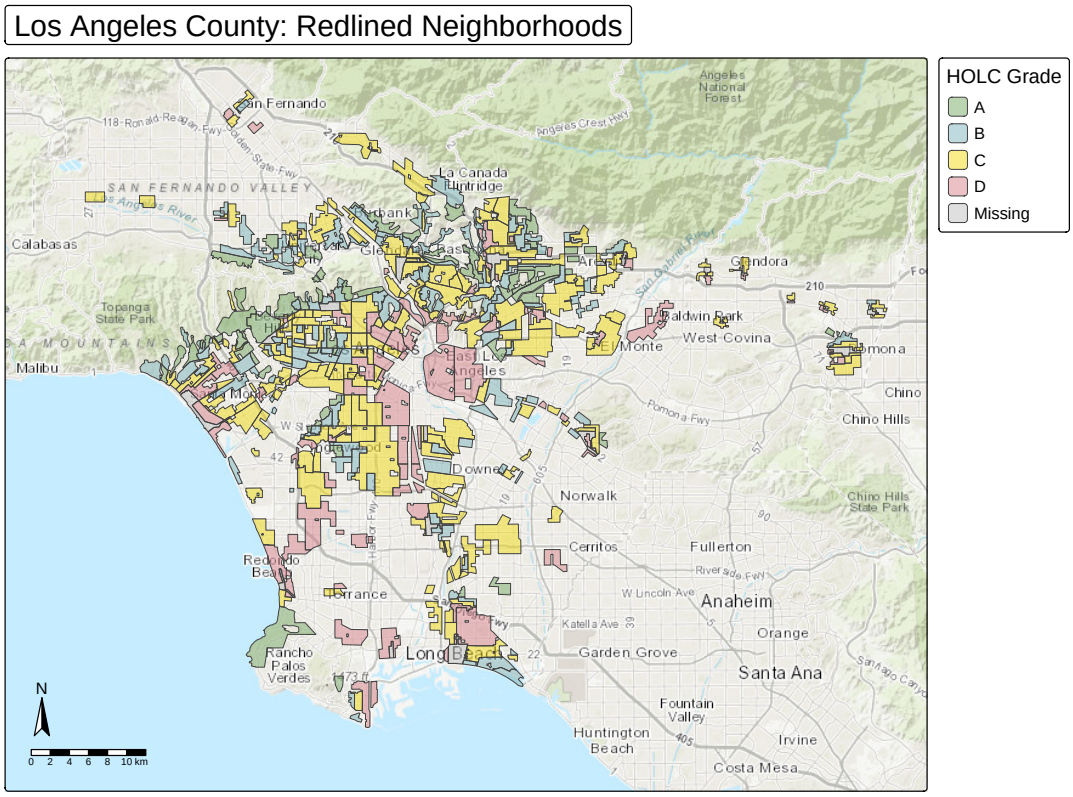


Figure 1: Map of Los Angeles (LA) county in California (CA). Neighborhoods with HOLC grades assigned from 1935-1940 - A, B, C, and D - are represented on this map by color. An "A" HOLC grade is the best "residential security", while a "D" HOLC grade is the worst "residential security."

Question 2: Table of Census Blocks Within HOLC Grades

```
# Filter ejscreen to los angeles county in CA
los_angeles_ejs <- ejscreen |>
    dplyr::filter((ST_ABBREV == "CA") &
                (CNTY_NAME == "Los Angeles County"))

# Confirm st_intersect will work in st_join for ejscreen LA and holc datasets
if(any(st_intersects(los_angeles_ejs, holc, sparse = FALSE))){
  print("st_intersect will work - geometries overlap.")
} else{
```

```
  stop("st_intersect will not work - no overlapping geometries.")
}
```

```
[1] "st_intersect will work - geometries overlap."
```

```
# Join holc to los_angeles_ejs by intersecting geometry
la_holc <- st_join(los_angeles_ejs, holc, join = st_intersects) |>
    # Select columns relevant to Part 1
    select(ID,grade,LOWINCPCT,P_PM25,P_LIFEEXPPCT) |>
    # Drop geometries
    st_drop_geometry()

# Warning - Join - check if number of rows matches original data (left join)
if(nrow(los_angeles_ejs) == nrow(la_holc)){
  print("Join matches original data dimensions of los_angeles_ejs")
} else{
  warning("Join does not match original data dimensions of los_angeles_ejs.")
  print(paste("los_angeles_ejs has", nrow(los_angeles_ejs), "rows"))
  print(paste("la_holc has", nrow(la_holc), "rows"))
  print("Duplicated rows are census blocks intersected by 2+ grades.")
  print("Duplicates will be kept - see Results Summary for details.")
}
```

```
Warning: Join does not match original data dimensions of los_angeles_ejs.
```

```
[1] "los_angeles_ejs has 6591 rows"
[1] "la_holc has 8988 rows"
[1] "Duplicated rows are census blocks intersected by 2+ grades."
[1] "Duplicates will be kept - see Results Summary for details."
```

```
# Create summary table of census block (cb) counts and percentages per holc grade
la_holc_graden <- la_holc |>
    group_by(grade) |>
    summarise(cb_count = n()) |>
    mutate(cb_perc = (cb_count / nrow(la_holc)) * 100)

# Use kable to format table output
kable(la_holc_graden,
        col.names = c("HOLC Grade","Census Blocks per Grade (Count)",
                      "Census Blocks per Grade (Percent)"))
```

Table 1: Counts and percentages of EJScreen census blocks per HOLC Grade in LA county. The percentages were calculated using a total LA county census block total of 8988 (see Results Summary for details).

HOLC Grade	Census Blocks per Grade (Count)	Census Blocks per Grade (Percent)
A	449	4.99555
B	1239	13.78505
C	3058	34.02314
D	1346	14.97552
NA	2896	32.22074

Question 3: EJScreen Data Visualizations Within HOLC Grades

```
# Create df for visualizations
la_holc_sum <- la_holc |>
    # Group by grade
    group_by(grade) |>
    # Calculate means by grade
    summarise(mean_LOWINCPCT = mean(LOWINCPCT, na.rm = TRUE),
              mean_P_PM25 = mean(P_PM25, na.rm = TRUE),
              mean_P_LIFEEXPPCT = mean(P_LIFEEXPPCT, na.rm = TRUE)) |>
    rename("HOLC Grade" = "grade")
```

```
# Generate plots

# Generate bar plot - mean_LOWINCPCT across HOLC grades
sum_plot1 <- ggplot(la_holc_sum,
    aes(x = `HOLC Grade`,
        y = mean_LOWINCPCT*100,
        fill = `HOLC Grade`)) +
    geom_bar(stat = "identity") +
    scale_fill_manual(values = holc_colors) +
    theme(axis.title.x = element_blank(),
          axis.text.x  = element_blank()) +
    labs(title = "Low Income",
         y = "Mean %")

# Generate bar plot - mean_P_PM25 across HOLC grades
sum_plot2 <- ggplot(la_holc_sum,
```

```
    aes(x = `HOLC Grade`,
        y = mean_P_PM25,
        fill = `HOLC Grade`)) +
    geom_bar(stat = "identity") +
    scale_fill_manual(values = holc_colors) +
    theme(axis.title.x = element_blank(),
          axis.text.x  = element_blank()) +
    labs(title = "Particulate Matter 2.5",
         y = "Mean US Percentile")

# Generate bar plot - mean_P_LIFEEXPPCT across HOLC grades
sum_plot3 <- ggplot(la_holc_sum,
    aes(x = `HOLC Grade`,
        y = mean_P_LIFEEXPPCT,
        fill = `HOLC Grade`)) +
    geom_bar(stat = "identity") +
    scale_fill_manual(values = holc_colors) +
    theme(axis.title.x = element_blank(),
          axis.text.x  = element_blank()) +
    labs(title = "Life Expectancy",
         y = "Mean US Percentile")

# Arrange bar plots in one figure
ggarrange(sum_plot1, sum_plot2, sum_plot3, ncol = 3, nrow = 1,
            labels = c("(a)", "(b)", "(c)"),
            font.label = list(face = "plain"),
            common.legend = TRUE,
            legend = "bottom") |>
# Add figure title
annotate_figure(top = text_grob("HOLC Neighborhoods: Current EJScreen Statistics",
                                size = 20))
```

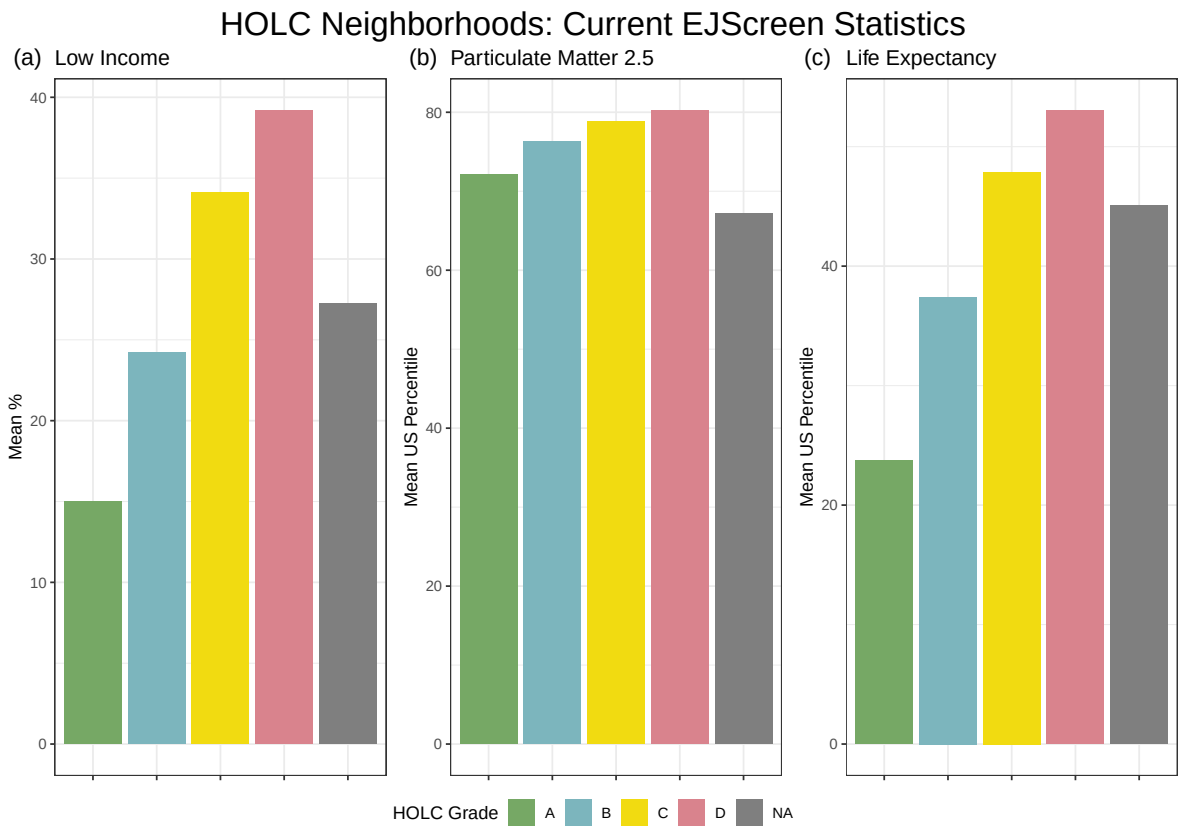


Figure 2: Bar plots showing current (2023) EJScreen data for three statistics - (a) low income (%), (b) US percentile for Particulate Matter (PM) 2.5, and (c) US percentile for low life expectancy - each averaged across historical HOLC neighborhood (redlining) grades.

Question 4: Results Summary

For LA county, the number of EJScreen census blocks within a Home Owners' Loan Corporation (HOLC) grade is not evenly distributed across the four grades - "A", "B", "C", "D". The "A" and "B" grades combined only have around 19% of the total LA county census blocks noted in EJScreen, while grade "C" has about 34% and grade "D" has 15% (32% of the census blocks were not associated with a grade) (*Table 2*). It should be noted that the total number of EJScreen census blocks in LA county was 6591, this number was increased to 8988 due to 1731 cases of multiple HOLC grades appearing in a single census block. To preserve HOLC data, data from a single census block may be used in calculating summary statistics for more than one HOLC grade.

When comparing the HOLC grade means of three metrics recorded per census block in EJScreen - low income (%), US percentile for Particulate Matter (PM) 2.5, and US percentile for low life expectancy - a similar trend was observed across all three comparisons. The means calculated

for low income (%), percentile for PM2.5, and percentile for low life expectancy all increase sequentially from grade "A" to grade "D" (*Figure 2*). This implies that historical redlining by HOLC from 1935-1940 is possibly correlated with current (2023 EJScreen) trends in low income, PM2.5, and low life expectancy. In other words, the census blocks located in grade "A" HOLC neighborhoods currently have the lowest mean percentage of low income households (14.89%), and fall in lower mean percentiles than the other three grades for annual PM2.5 air exposure (~72th percentile) and low life expectancy (~24th percentile). In contrast, grade "D" HOLC neighborhoods have the highest mean percentage of low income households (39.19%), and fall in higher mean percentiles than the other three grades for annual PM2.5 air exposure (~80th percentile) and low life expectancy (~53rd percentile). An alternative interpretation of the US percentile metric is that if a group - like grade "D" - is in the 53rd US percentile for low life expectancy, then only 47% of the US has a higher low life expectancy metric. Given these results, it appears that historically redlined "C" and "D" grade census blocks do currently experience more negative environmental and socioeconomic conditions than census blocks in historical HOLC "A" and "B" grade neighborhoods.

Part 2: Legacy of redlining in pedestrian and bike accidents

Your second task is to explore the legacy of historical redlining in Los Angeles on the report of pedestrian and bike crashes/ fatalities.

For this assignment, you must produce the following based on observations from 2020-2025:

1. A figure summarizing the percent of observations within redlined neighborhoods within each HOLC grade
    - Create a visualization that shows:
    - The percentage of bike and pedestrian crashes and fatalities within each HOLC grade
    - Include an appropiate title, axis labels, and legend

```
# Filter crashes to only Y PED accidents
ped_crashes_filt <- crashes |>
    # Select only relevant columns to Part 2 (geometry sticks)
    select(CASE_ID, PEDESTRIAN_ACCIDENT, COLLISION_SEVERITY) |>
    # Apply filter
    filter(PEDESTRIAN_ACCIDENT == "Y")

# Filter crashes to only Y BIKE accidents
bike_crashes_filt <- crashes |>
    # Select only relevant columns to Part 2 (geometry sticks)
    select(CASE_ID, BICYCLE_ACCIDENT, COLLISION_SEVERITY) |>
    # Apply filter
```

```
    filter(BICYCLE_ACCIDENT == "Y")

# Confirm st_intersect will work in st_join for holc and crashes_filt datasets
if((any(st_intersects(holc, ped_crashes_filt, sparse = FALSE))) &
    (any(st_intersects(holc, bike_crashes_filt, sparse = FALSE)))){
  print("st_intersect will work - geometries overlap.")
} else{
  stop("st_intersect will not work - no overlapping geometries.")
}
```

```
[1] "st_intersect will work - geometries overlap."
```

```
# Join holc to ped_crashes by intersecting geometry
holc_ped_crash <- st_join(ped_crashes_filt, holc, join = st_intersects) |>
    # Select columns relevant to Part 2
    select(grade, CASE_ID, PEDESTRIAN_ACCIDENT, COLLISION_SEVERITY) |>
    # Drop geometries
    st_drop_geometry()

# Join holc to bike_crashes by intersecting geometry
holc_bike_crash <- st_join(bike_crashes_filt, holc, join = st_intersects) |>
    # Select columns relevant to Part 2
    select(grade, CASE_ID, BICYCLE_ACCIDENT, COLLISION_SEVERITY) |>
    # Drop geometries
    st_drop_geometry()

# Check for duplicates - none, all IDs have 1 grade match
ped_crash_flag <- holc_ped_crash |>
    group_by(CASE_ID) |>
    summarise(Num_Grade_matches = n()) |>
    arrange(Num_Grade_matches, ascending = FALSE)
bike_crash_flag <- holc_bike_crash |>
    group_by(CASE_ID) |>
    summarise(Num_Grade_matches = n()) |>
    arrange(Num_Grade_matches, ascending = FALSE)

# Warning - Join - check if number of rows matches original data (left join)
if((nrow(holc_ped_crash) == nrow(ped_crashes_filt)) &
    (nrow(holc_bike_crash) == nrow(bike_crashes_filt))){
  print("Joins match data dimensions of ped_crashes_filt and bike_crashes_filt.")
} else{
```

```
  warning("Joins do not match original data dimensions.")
}
```

```
[1] "Joins match data dimensions of ped_crashes_filt and bike_crashes_filt."
```

```
# Create ped summary df
sum_ped_crash <- holc_ped_crash |>
    # Group by HOLC grade
    group_by(grade) |>
    summarise(
        # Count ped accidents per grade
        num_Ped_Acc = n(),
        # Percent ped accidents per grade
        perc_Ped_Acc = num_Ped_Acc/nrow(holc_ped_crash) * 100,
        # Count total fatalities per grade
        # Each row where COLLISION_SEVERITY == 1.00 is TRUE equals 1
        # Sum those 1s to get a count of rows with a value of TRUE
        num_Ped_Fatal = sum(COLLISION_SEVERITY == 1.00),
        # Percent total fatalities per grade
        perc_Ped_Fatal = num_Ped_Fatal/nrow(holc_ped_crash) * 100
    ) |>
    rename(`HOLC Grade` = grade)

# Create bike summary df
sum_bike_crash <- holc_bike_crash |>
    # Group by HOLC grade
    group_by(grade) |>
    summarise(
        # Count total ped accidents per grade
        num_Bike_Acc = n(),
        # Percent ped accidents per grade
        perc_Bike_Acc = num_Bike_Acc/nrow(holc_bike_crash) * 100,
        # Count total fatalities per grade
        # Each COLLISION_SEVERITY == 1.00 is TRUE = 1
        # Sum those 1s to get a count of rows with a value of TRUE
        num_Bike_Fatal = sum(COLLISION_SEVERITY == 1.00),
        # Percent total fatalities per grade
        perc_Bike_Fatal = num_Bike_Fatal/nrow(holc_bike_crash) * 100
    ) |>
    rename(`HOLC Grade` = grade)
```

```r
# Plot visualization

# Plot 1: Pedestrian Accidents
plt1 <- ggplot(sum_ped_crash,
            aes(x = `HOLC Grade`,
                y = perc_Ped_Acc,
                fill = `HOLC Grade`)) +
            geom_bar(stat = "identity") +
            scale_fill_manual(values = holc_colors) +
            theme(axis.title.x = element_blank(),
                  axis.text.x  = element_blank()) +
            ylim(0,60) +
            labs(title = "Pedestrian Accidents",
                 y = "Accidents (%)")

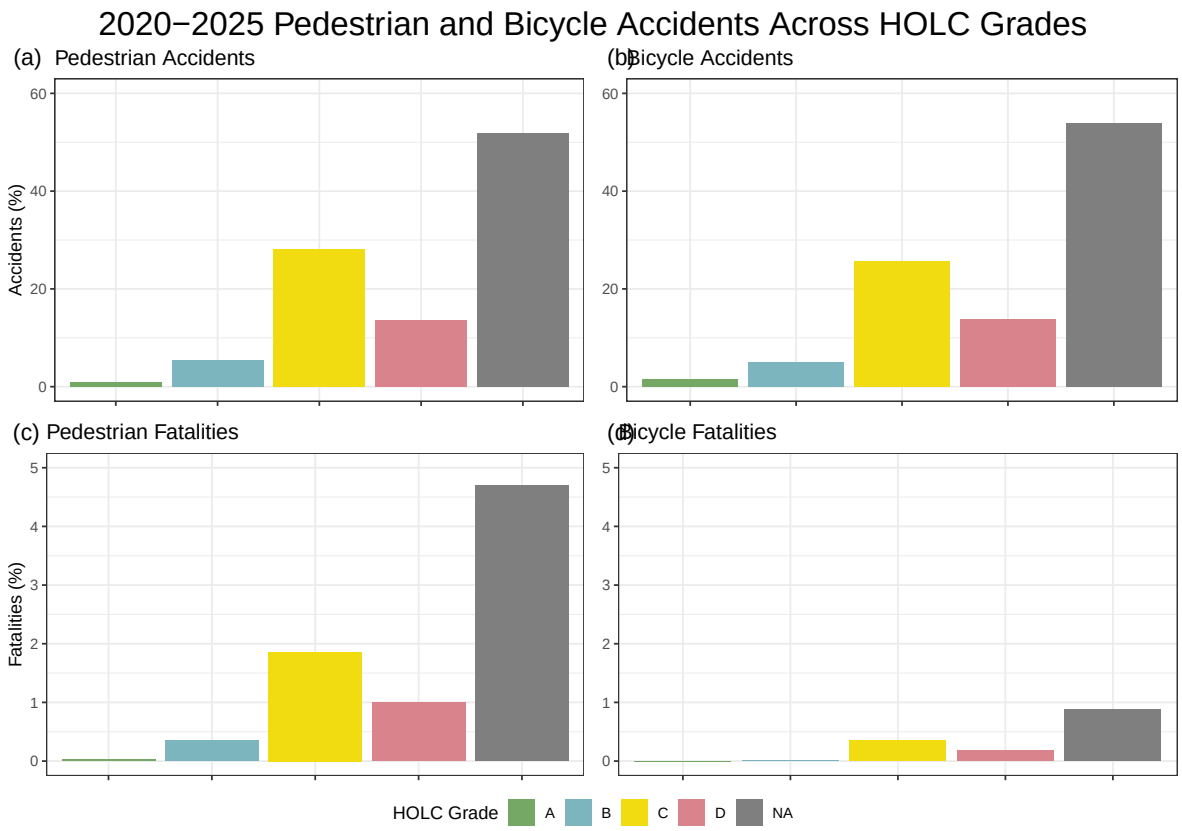
# Plot 2: Bike Accidents
plt2 <- ggplot(sum_bike_crash,
            aes(x = `HOLC Grade`,
                y = perc_Bike_Acc,
                fill = `HOLC Grade`)) +
            geom_bar(stat = "identity") +
            scale_fill_manual(values = holc_colors) +
            theme(axis.title.x = element_blank(),
                  axis.text.x  = element_blank(),
                  axis.title.y  = element_blank()) +
            ylim(0,60) +
            labs(title = "Bicycle Accidents")

# Plot 3: Pedestrian Accidents
plt3 <- ggplot(sum_ped_crash,
            aes(x = `HOLC Grade`,
                y = perc_Ped_Fatal,
                fill = `HOLC Grade`)) +
            geom_bar(stat = "identity") +
            scale_fill_manual(values = holc_colors) +
            theme(axis.title.x = element_blank(),
                  axis.text.x  = element_blank()) +
            ylim(0,5) +
            labs(title = "Pedestrian Fatalities",
                 y = "Fatalities (%)")

# Plot 4: Bike Accidents
```

```r
plt4 <- ggplot(sum_bike_crash,
            aes(x = `HOLC Grade`,
                y = perc_Bike_Fatal,
                fill = `HOLC Grade`)) +
            geom_bar(stat = "identity") +
            scale_fill_manual(values = holc_colors) +
            theme(axis.title.x = element_blank(),
                  axis.text.x  = element_blank(),
                  axis.title.y  = element_blank()) +
            ylim(0,5) +
            labs(title = "Bicycle Fatalities")

# Arrange bar plots in one figure
ggarrange(plt1, plt2, plt3, plt4, ncol = 2, nrow = 2,
            labels = c("(a)", "(b)", "(c)", "(d)"),
            font.label = list(face = "plain"),
            common.legend = TRUE,
            legend = "bottom") |>
# Add figure title
annotate_figure(
    top = text_grob("2020-2025 Pedestrian and Bicycle Accidents Across HOLC Grades",
                    size = 20))
```



Data

The data used for this assignment is from the EJScreen: Environmental Justice Screening and Mapping Tool, which no longer exists online. However, our data can be accessed through the following link: https://drive.google.com/file/d/1nG6Nj1bXfzQFOVMO8Km3eNy4SWu1YcIQ/view

References

EDS223 - UCSB MEDS 2027